

StegoRSA

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QUESTION

StegoRSA

Cryptography **Easy**

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A message has been encrypted using RSA. The public key is gone... but someone might have been careless with the private key. Can you recover it and decrypt the message?

Download the [flag](#) ↗ and [image](#) ↗.

HINTS ⓘ

1

Metadata can tell you more than you expect.

2

Hex can be turned back into a key file.

FLAG ⓘ

Submit

Metadata from the image file

```
(kali@kali)-[~/ctf/stegorsa]
└─$ exiftool image.jpg
ExifTool Version Number      : 13.50
File Name                    : image.jpg
Directory                    : .
File Size                    : 21 kB
File Modification Date/Time   : 2026:02:07 10:26:34-05:00
File Access Date/Time        : 2026:07:06 10:00:07-04:00
File Inode Change Date/Time   : 2026:07:06 09:59:43-04:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
Resolution Unit               : None
X Resolution                  : 1
Y Resolution                  : 1
Comment                      : 2d2d2d2d2d424547494e2050524956415445204b45592d2d2d2d0a4d494945766749424144414e42676b71686b694739773042415145464141534342
4b67776767536b41674541416f494241514330347a793268776946584944680a5072776b623272f2493047352b316465567a4468465359654b655576467785a50733777724f303969343959683355
6e5565364a76335a2b72434e357a2b5965330a78674c336938437647494b45534e3172685a474b4b676e5a714c656c694e4c385139326547514353524a6c556677316b5845523578716951304856
67503946430a77325736385a42437a3875447054782b36385736584e367246752f4f4b31566c776c3170584365555675504c5a57634c4a2f34682f4f4d6a4b4e62325362676b0a31597636464768
6a4a714a6165505a456a675135567346646c6756546f4a2b6371364638797a7a5261347077425a683259516242584f674a3875655976656b470a62466e5935537130327152445577526142354d33
375255784662786566373249516e59435a43794b73315264727a49394e764d7a53566c314734545776675a480a46534f6653506a2f41674d42414145436767454142426732756830425a4e397372
57795a6e71754642454c636a49537a664265324c416f7142446e486b464b300a3052594d39742b5055766f38555347374476f654362453347597949574a43338554743744b374e72456a2b5031
645535386573765464705a4b7473694d73350a6c327444323334f5a5145336856476738486f6b4e4b7436436c6b6c70776d617834756278476b4f626d79387578796b363279543237325a627238
4571364b470a736f514f3474736632614a4873704d6e356232587534656f3672436c33623341324e4d6c47324659444152524e65326862555744565241776d446552456171660a595979754f6173
4a4a7577326766754555644e3547305a635947736c5a6a62757177614930447a63723434455566566534f584533637641486968636a6e4d610a6659496d43365242394478333442706656756855
5a6a7374316161516644426e4b566554595a675239514b426751444567527a517a543661596b37304d366f330a326f4e5a56544547742f7534387a76345a44685448526966353265584e47347675
446a64537744517a31465a6d66365a4956484747592b553173464f6d6656770a636b3236363714b32556736344e385237346b65517444336f7a4a4e7537704b30464142364c416250533771472b
722b5a486347306e5963376e4854753668620a4677747351535a52764171357152476865534541705a494f66514b426751447270362f72695175527555796c5642787437327053537657336d6870
4d523068520a304b6e396e586d6775792f797344557154727a435a6865487652642b3534434b486f41656449666635626939766d72666b4e5a366b48734b702b65314d7133770a4e484b54582f4d
5569684b6f79727671747932366f4b73736837542f32597733445a696179386f38764c596a46796650612f462b4446707067714f4a597a41570a4b4b364e373762534b774b42675143745439726b
5a6b4753314b5969522f635863416874482f417670744844572f52477263336e73442f70754b54712f586b350a61566675425a72676b426d4631345a76394b3030395a4a6e69636e51303456662b
6443497a47367861396143353561584d5264557330577652436133334a30410a4978364e73486179623549544e54426230717751506b7569564d50634a53546c776a74323754636f323572794353
58647277486d669467032514b426751436b0a674a46676c5846624434512b6f534762614a624b7a594d4c344f706374794833347662586b5064587958506d4d5456627a45306555314133744474
546e6c31760a6a5730595658614953482b584450626b432b4a382b714335786f344d516b324473636b4e73647770535631504d503750455949396c6d30486a394139446574410a667941355a7574
4e544e7245556953374f666f42614c51776c543539364a796e793078577a72444c73774b42674573466f2f2b76346d665356646c396551326c0a324265712f6d6543786a5732685453562786b664f
4d576379746a774a327259462f586c4a6973355a5a537345574e307570474150486a65684a503763494642710a4f394155386d4c715a4f42394d3638327646386452462f4d44f5871614e4f39492f
3474774b6f36396a384c442f32676b725951496c4e72396d663644527a324b0a676870637250383071526c4e30524d6930704a536966782b0a2d2d2d2d454e442050524956415445204b45592d
2d2d2d2d0a
Image Width                  : 512
Image Height                 : 512
Encoding Process             : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 512x512
Megapixels                   : 0.262
```

Convert the commented hex part to raw value

```

-----BEGIN PRIVATE KEY-----
MIIEvGIBADANBqkqhkiG9w0BAQEFAASCBAgEAAoIBAQC04zy2hwiFXIDh
Prwkb2w/I0G5+1deVzDhFSYeKeWdgxZPs7wr009i49Yh3UnUe6Jv3Z+rCN5z+Ye3
xgL3i8CvGIKESN1rhZGKKgnZqLeLiNL8Q92eGQCSRJlUfw1kXER5xqiQ0HVgP9FC
w2W68ZBCz8uDpTx+68W6XN6rFu/OK1Vlw1pXCeUVuPLZWcLJ/4h/OMjKNb2Sbgk
1Yv6FGhjJqJaePZEjgQ5VsFd1gVToJ+cq6F8yzzRa4pwBZh2YQbBX0gJ8ueYvekG
bFnY5Sq02qRDUwRaB5M37RUxFbxef72IQnYCZCyKs1RdrzI9NvMzSV11G4TWvgZH
FS0fSPj/AgMBAAECggEABBg2uh0BZN9srWyZnquFBELcjISzfBe2LAoQBDnHkFK0
0RYM9t+PUvo8USTstGoeCbE3GYyIWJC38UGCtK7NrEj+P1dU58esvTdpZKtsiMs5
12tD2330ZQE3hVGg8HokNKt6Clklpwwmax4ubxGk0bmy8uxyk62yT272Zbr8Eq6KG
soQ04tsf2aJHspMn5b2Xu4eo6rCl3b3A2NM1LrFYDARRNe2hbUWDVRAwmDeREaqf
YYyuOasJJuw2gfuEUdN5G0ZcYGs1ZjbuqwaI0Dzcr44EUkefSOXE3cvAHihcJnMa
fYImC6RB9Dx34BpfVuhUZjst1aaQfDBnKVeTYZgr9QKBgQDEgRzQzT6aYk70M6o3
2oNZVTEgt/u48zv4ZDhTHRif52eXNG4vuDjdSwDQz1FZmf6ZIVHGGY+U1sF0mfVw
ck266cqK2Ug64N8R74keQtD3ozJNu7pK0FAB6LabPS7qG+r+ZHcG0nYc7nHTu6hb
FwtsQSZRvAq5qRGheSEApZIOfQKBgQDrp6/riQuRuUy1VBxt72pSSvW3mhpMR0hR
0Kn9nXmguy/ysDUqTrzCZheHvRd+54CKHoAedIff5bi9vmrfkNZ6kHsKp+e1Mq3w
NHKTX/MUihKoyrvqty26oKssh7T/2Yw3DZiay8o8vLYjFyfPa/F+DFppgq0JYzAW
KK6N77bSKwKBgQCtT9rkZkGS1KYiR/cXcAhtH/AvptHDW/RGrc3nsD/puKTq/Xk5
aVfuBZrgkBmF14Zv9K009ZJnicnQ04Vf+dCIzG6xa9aC55aXMRdUs0WvRCa33J0A
Ix6NsHayb5ITNTBb0qwQPkuivMPcJSTlwjt27Tco25ryCSXdrwHmMiFp2QKBgQCK
gJFglXFbD4Q+oSGbaJbKzYML40pctyH34vbXkPdXyXPmMTVbzE0eU1A3tDtTn11v
jW0YVXaISH+XDPbkC+J8+qC5xo4MQk2DscKnsdwpSV1PMP7PEYI9lm0Hj9A9DetA
fyA5ZutNTNrEUiS70foBaLQw1T596Jyny0xWzrDLswKBgEsFo/+v4mfSVdl9eQ2l
2Beq/meCxjW2hT5bxkf0MWcytjwJ2rYF/X1Jis5ZZSsEWN0upGAPHjehJP7cIFBq
09AU8mLqZOB9M682vF8dRF/MOXqaN09I/4twKo69j8LD/2gkrYQI1Nr9mf6DRz2K
ghpcrP80qRlN0RMi0pJSifx+
-----END PRIVATE KEY-----

```

make this into a private file for decryption & use this command

```
openssl pkeyutl -decrypt -inkey privatekey -in flag.enc -out
decrypted_text.txt
```

openssl pkeyutl = the tool used for rsa decryption

-inkey = the file that needs to be decrypted.

-in = encrypted file

-out = the new file created that would have the decrypted value

```
(kali㉿kali)-[~/ctf/stegorsa]
$ openssl pkeyutl -decrypt -inkey privatekey -in flag.enc -out decrypted_text.txt

(kali㉿kali)-[~/ctf/stegorsa]
$ cat decrypted_text.txt
picoCTF{rs4_k3y_1n_1mg_66388eb3}
```